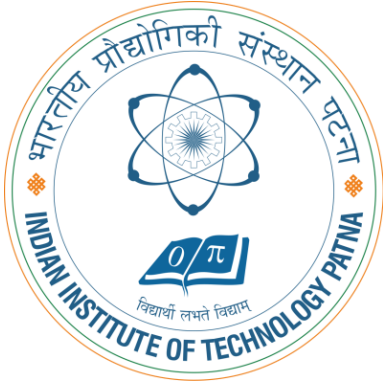


CS5102: FOUNDATIONS OF COMPUTER SYSTEMS

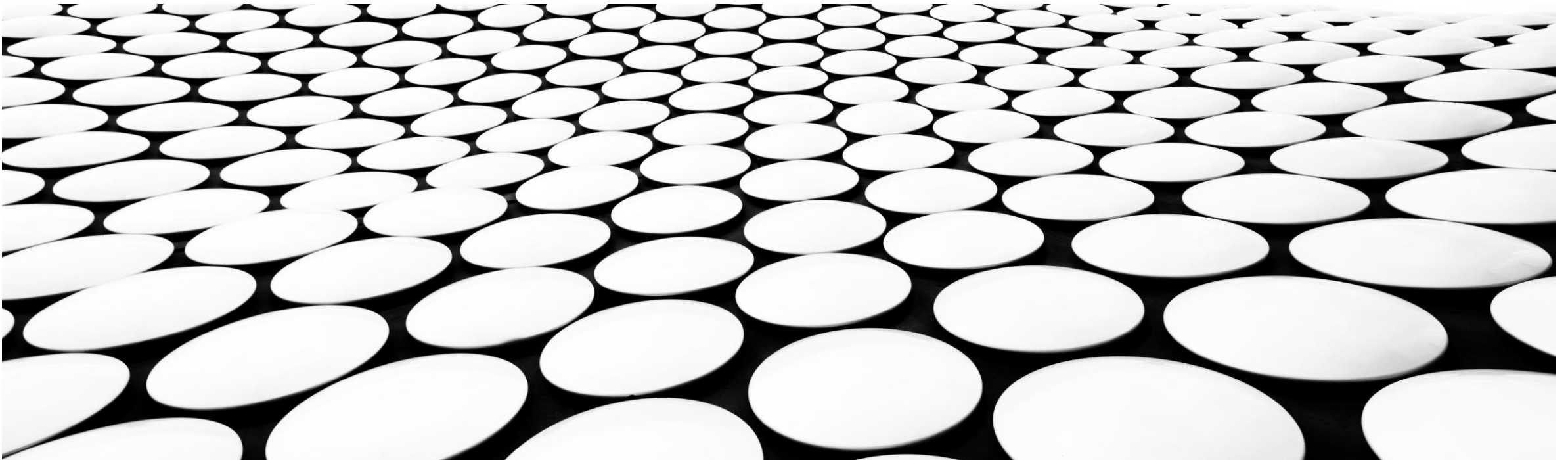


TOPIC-10: CONTROL UNIT DESIGN I

DR. ARIJIT ROY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY PATNA



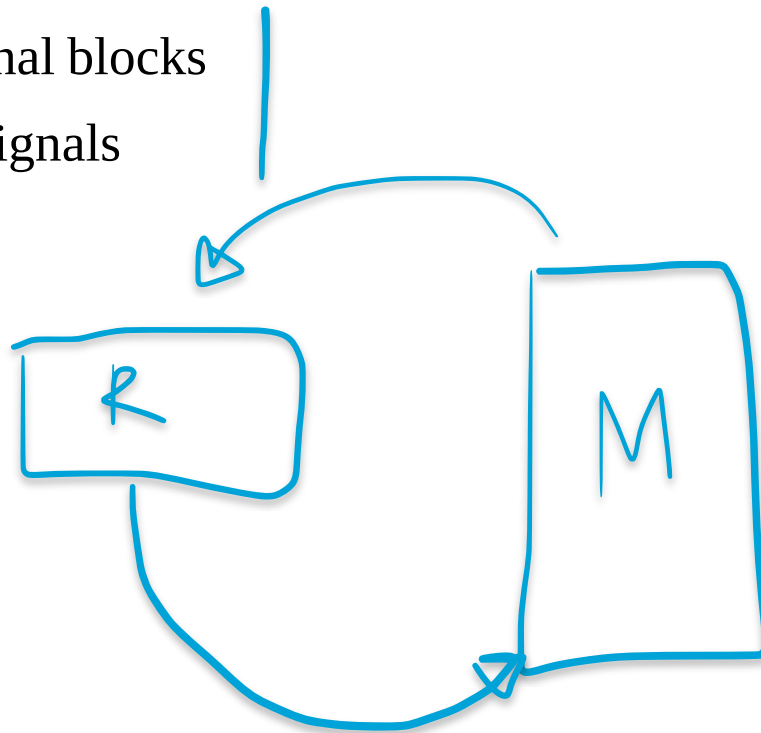
OVERVIEW



- Introduction
- Single-Cycle Processor ✓
- Performance Analysis ✓
- Multicycle Processor ✓
- Pipelined Processor ✓
- Exceptions
- Advanced Microarchitecture

INTRODUCTION

- Microarchitecture: how to implement an architecture in hardware
- Processor:
 - Datapath: functional blocks
 - Control: control signals



Application Software	programs
Operating Systems	device drivers
Architecture	instructions registers
Micro-architecture	datapaths controllers
Logic	adders memories
Digital Circuits	AND gates NOT gates
Analog Circuits	amplifiers filters
Devices	transistors diodes
Physics	electrons

MICROARCHITECTURE



- Multiple implementations for a single architecture:
 - Single-cycle
 - Each instruction executes in a single cycle
 - Multicycle ✓
 - Each instruction is broken up into a series of shorter steps
 - Pipelined ✓
 - Each instruction is broken up into a series of steps
 - Multiple instructions execute at once.

PROCESSOR PERFORMANCE

- Program execution time

$$\text{Execution Time} = \underbrace{(\# \text{ instructions})}_{\text{cycles/instruction}} \underbrace{(\text{seconds/cycle})}$$

- Definitions:

- Cycles/instruction = CPI ✓

- Seconds/cycle = clock period ✓

- $1/\text{CPI} = \text{Instructions/cycle} = \text{IPC}$ ✓

- Challenge is to satisfy constraints of:

- Cost

- Power

- Performance

MIPS PROCESSOR

- We consider a subset of MIPS instructions:
 - R-type instructions: and, or, add, sub, slt
 - Memory instructions: lw, sw
 - Branch instructions: beq
- Later consider adding addi and j

ARCHITECTURAL STATE



➤ Determines everything about a processor:

- ✓ ➤ PC
- ✓ ➤ 32 registers
- ✓ ➤ Memory ✓



SINGLE-CYCLE MIPS PROCESSOR

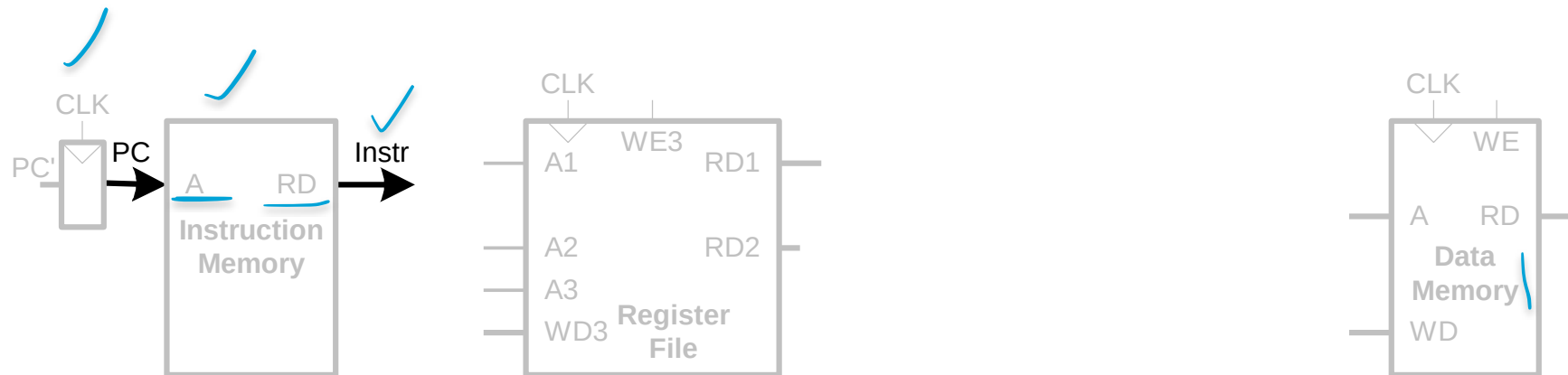


- Datapath
- Control

SINGLE-CYCLE DATAPATH: lw FETCH



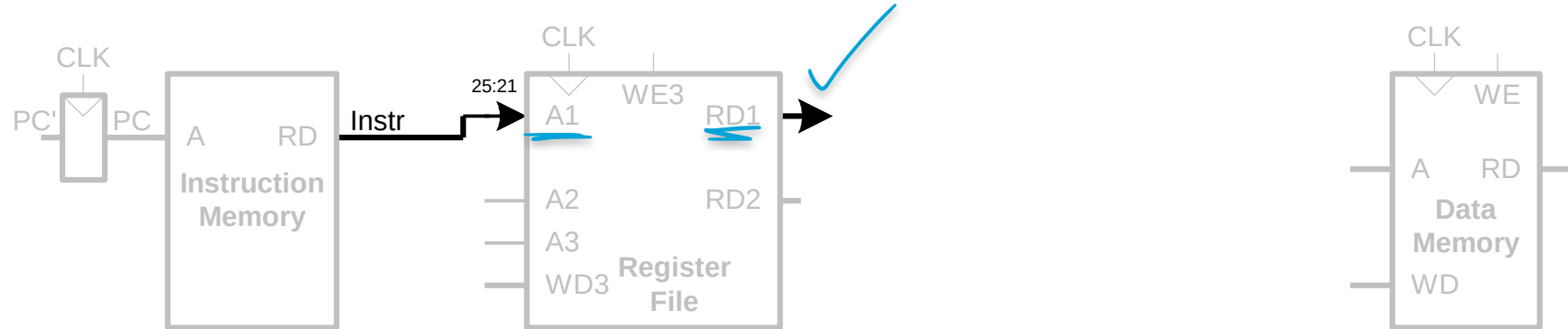
- First consider executing lw
- **STEP 1:** Fetch instruction



SINGLE-CYCLE DATAPATH: LW REGISTER READ

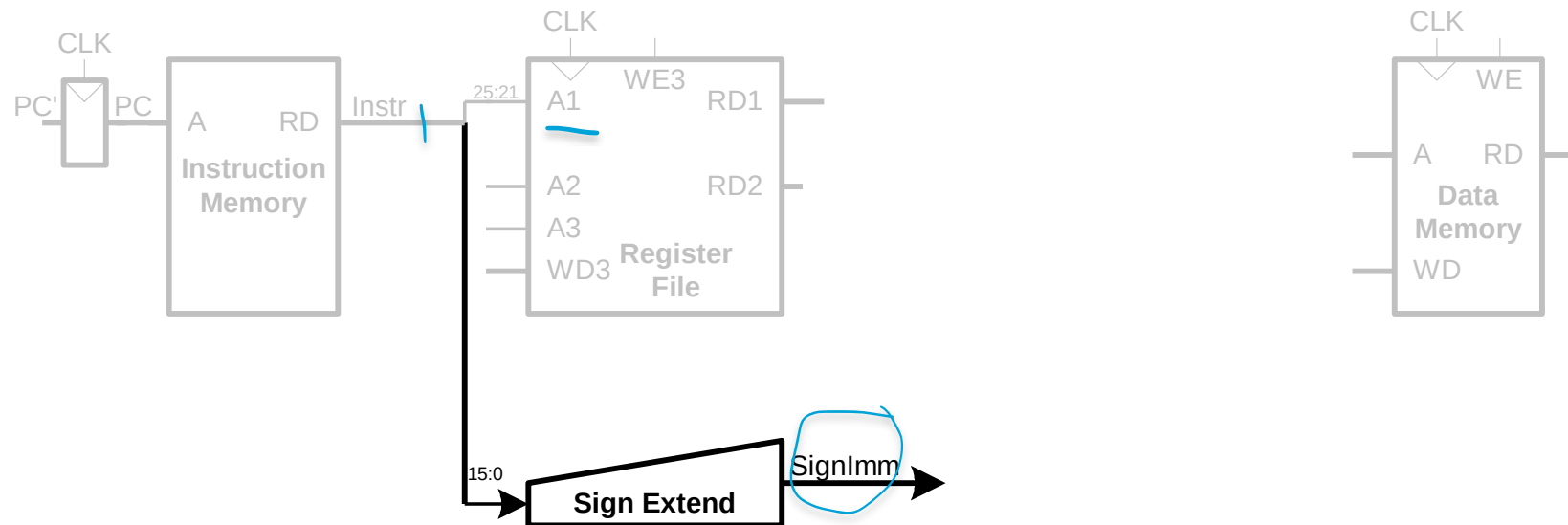


➤ **STEP 2:** Read source operands from register file



SINGLE-CYCLE DATAPATH: LW IMMEDIATE

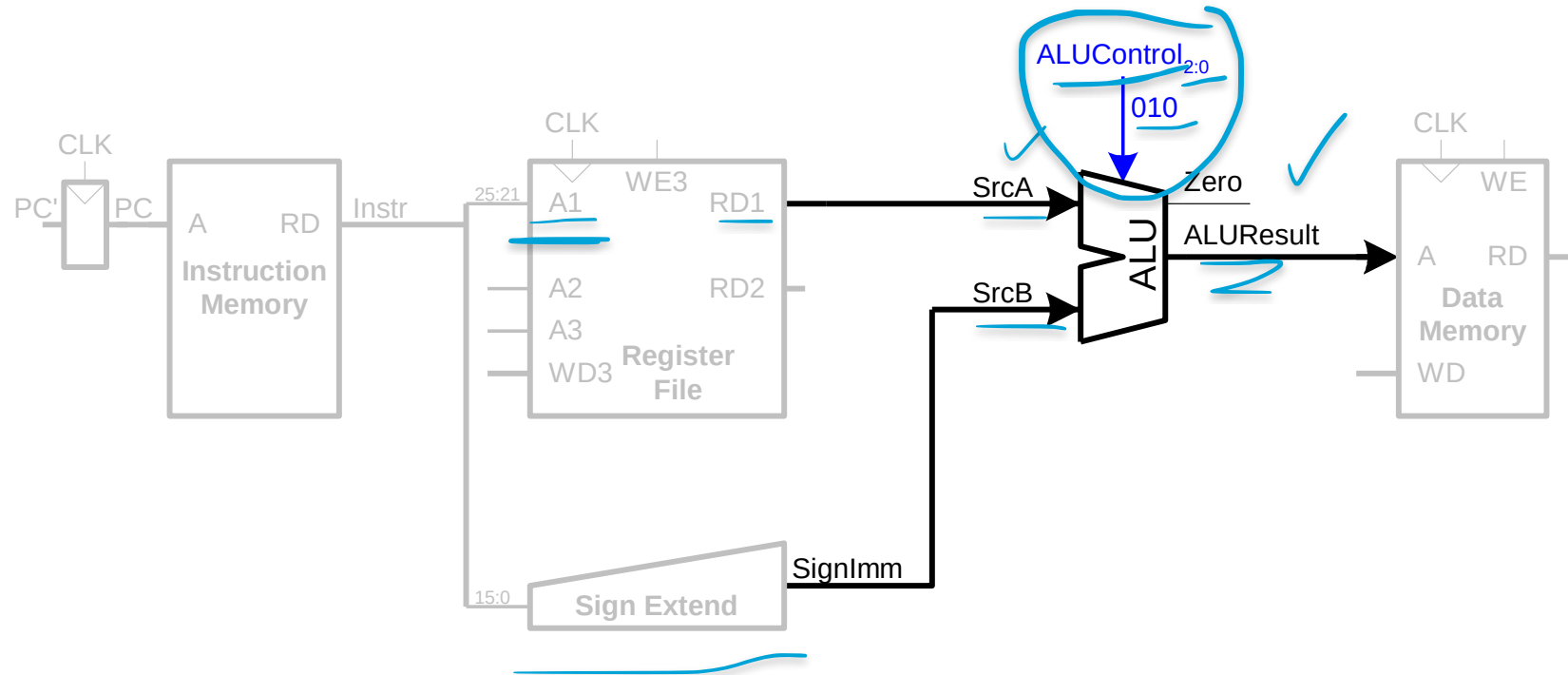
➤ STEP 3: Sign-extend the immediate



SINGLE-CYCLE DATAPATH: LW ADDRESS

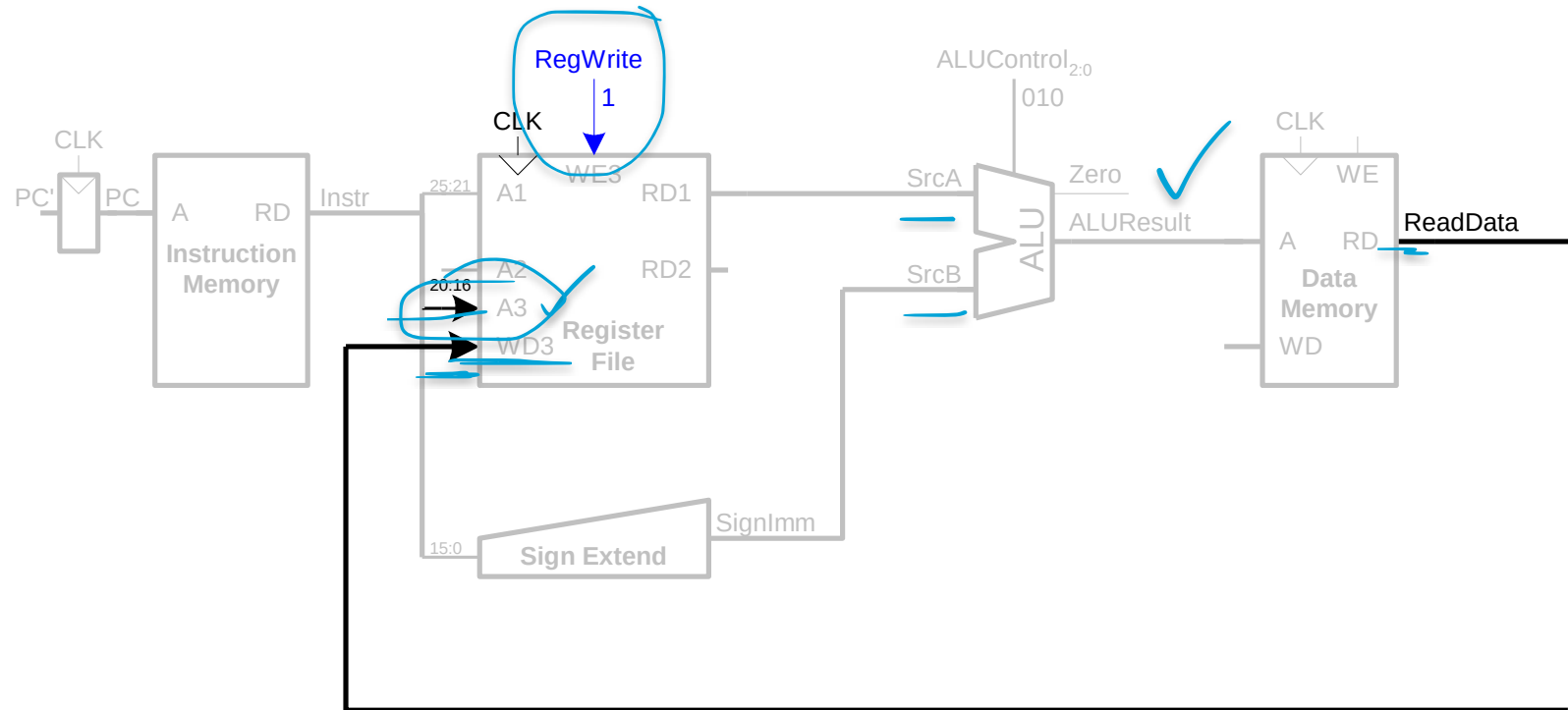
➤ **STEP 4:** Compute the memory address

\$20(4)



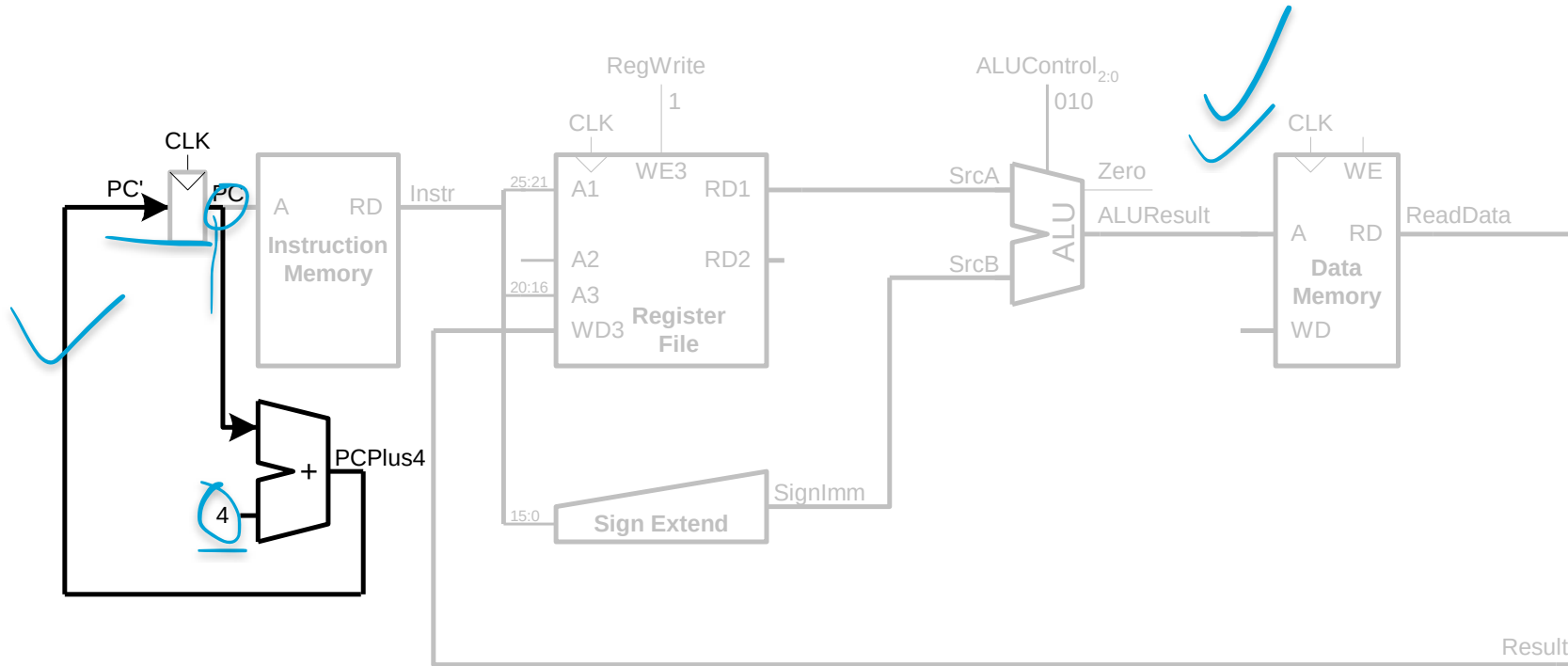
SINGLE-CYCLE DATAPATH: LW MEMORY READ

➤ **STEP 5:** Read data from memory and write it back to register file



SINGLE-CYCLE DATAPATH: LW PC INCREMENT

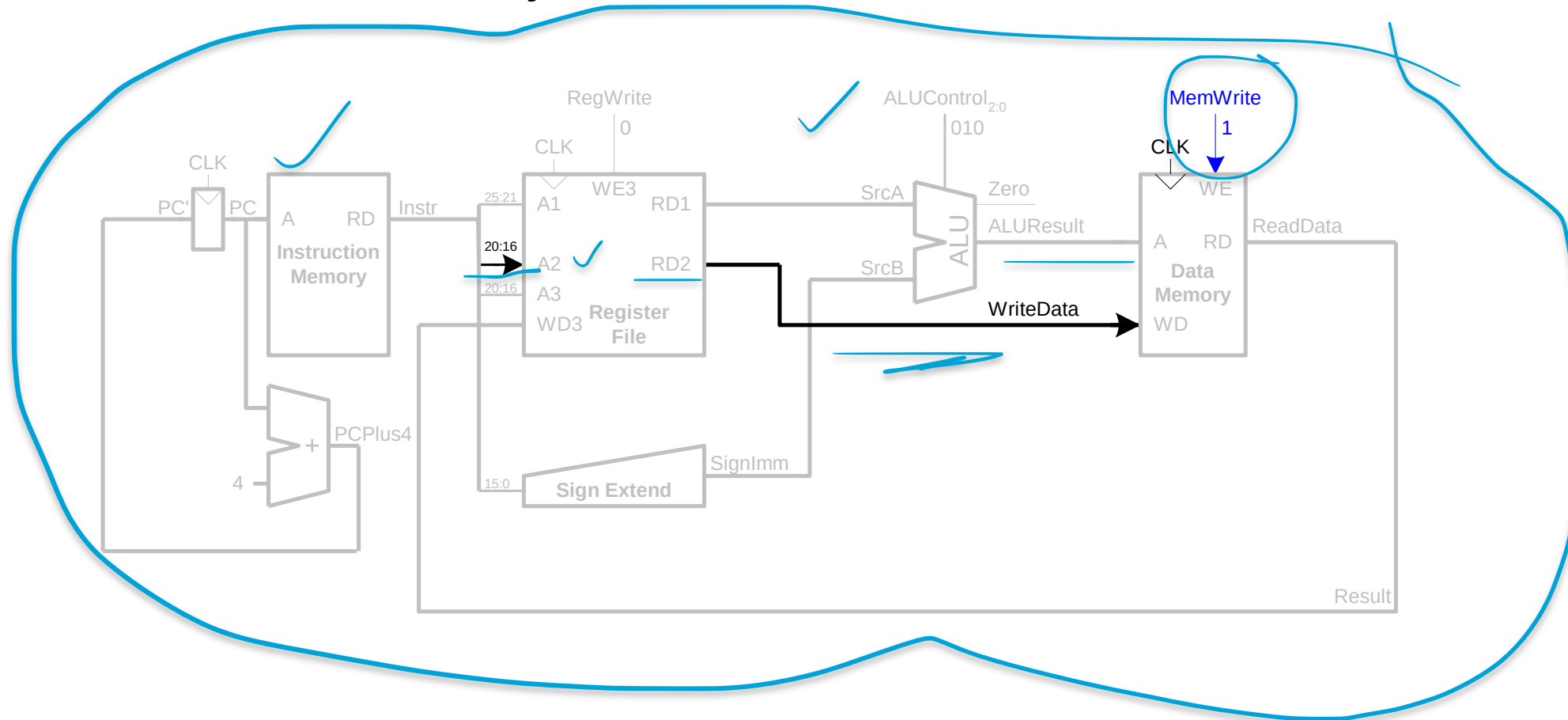
➤ **STEP 6:** Determine the address of the next instruction



SINGLE-CYCLE DATAPATH: SW



➤ Write data in rt to memory



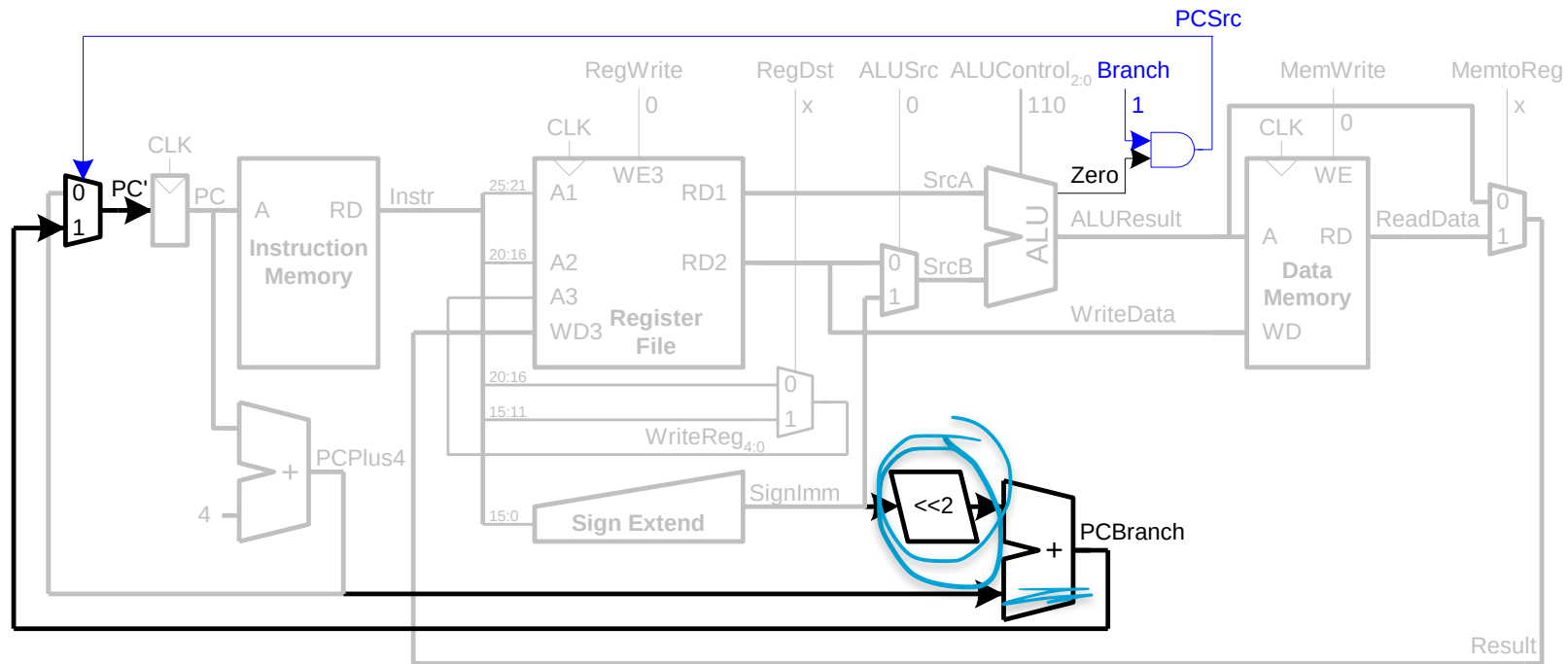
- Read from `rs` and `rt`
- Write ALU Result to register file
- Write to `rd` (instead of `rt`)



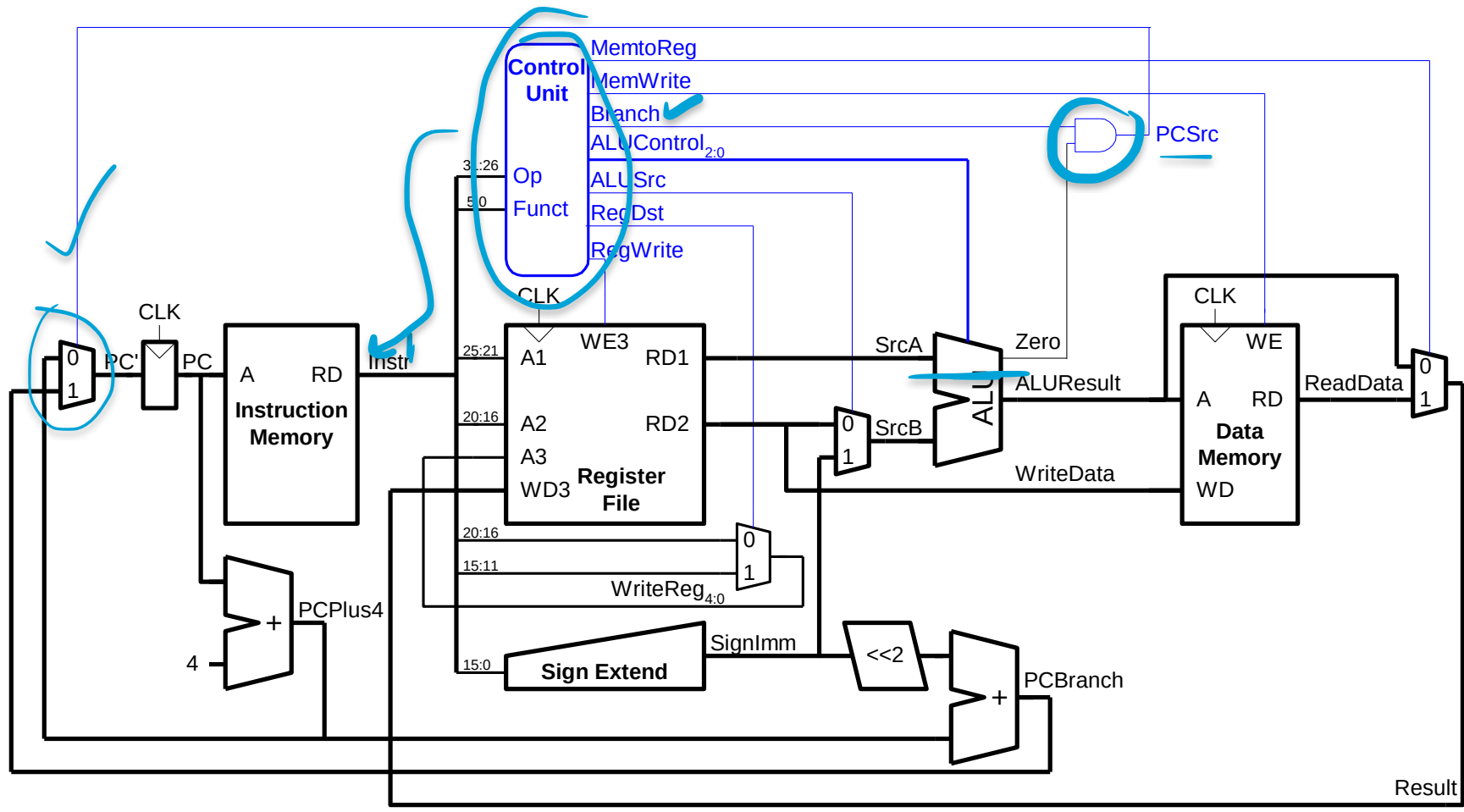
SINGLE-CYCLE DATAPATH: BEQ

- Determine whether values in rs and rt are equal
- Calculate branch target address:

$$\text{BTA} = (\text{sign-extended immediate} \ll 2) + (\text{PC} + 4)$$

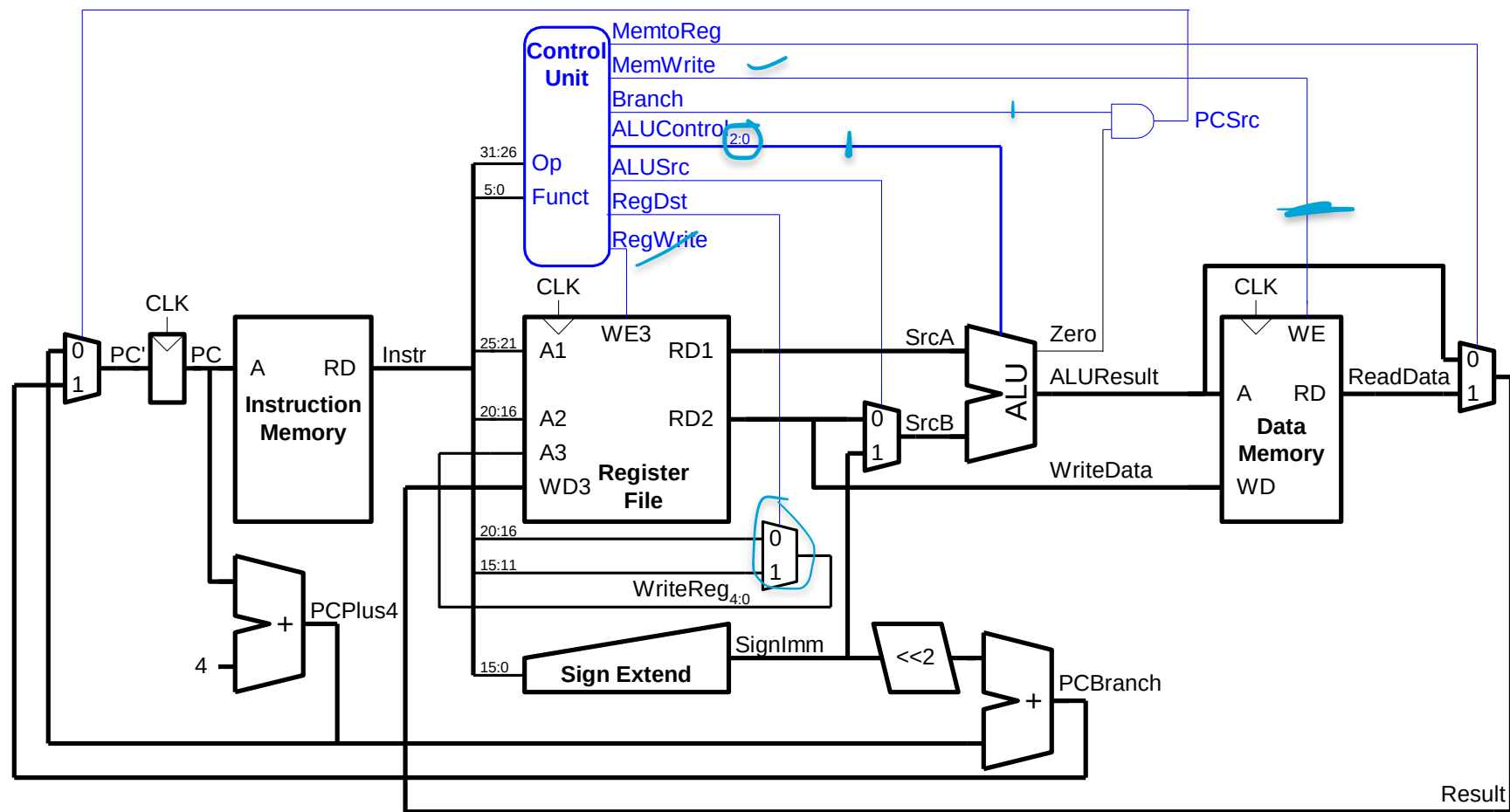


COMPLETE SINGLE-CYCLE PROCESSOR

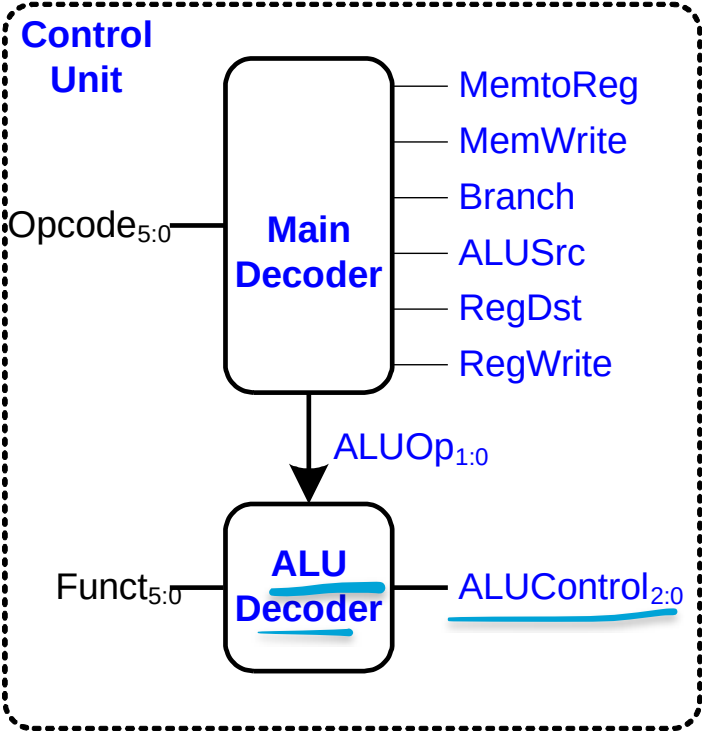


CONTROL SIGNALS

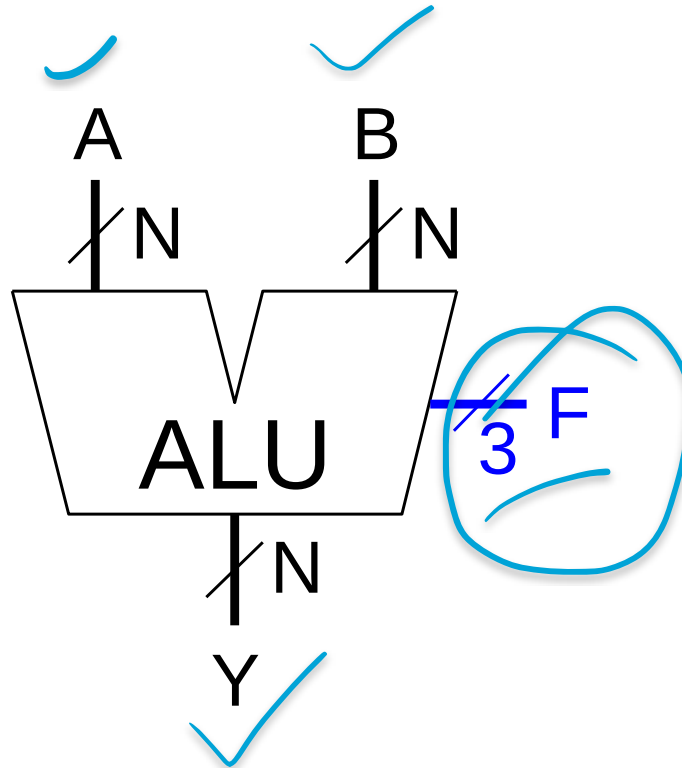
COMPLETE SINGLE-CYCLE PROCESSOR



CONTROL UNIT



REVIEW: ALU

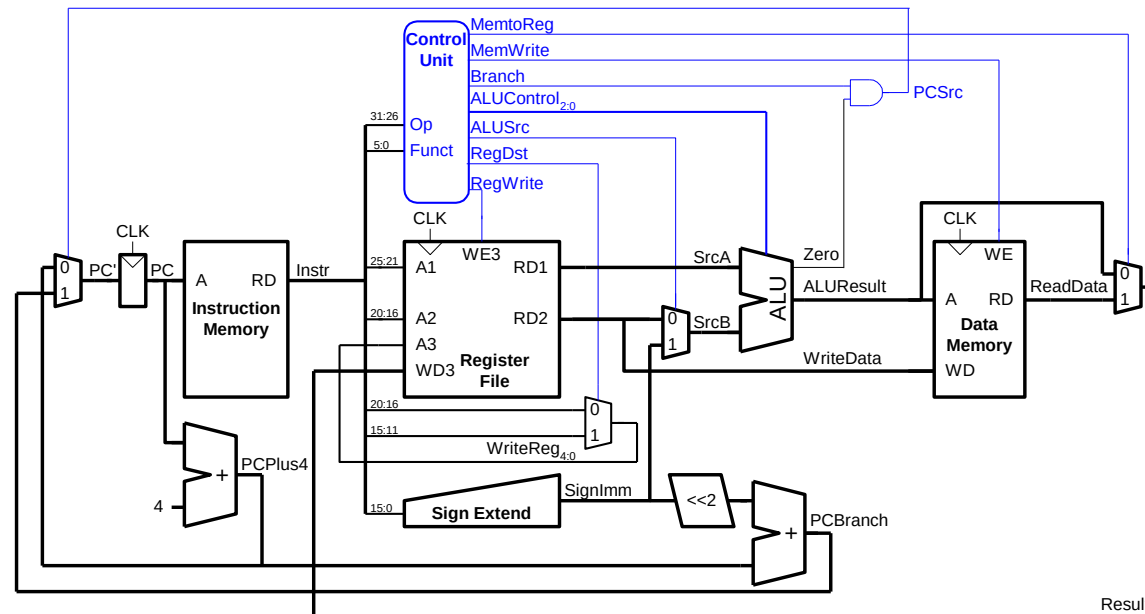


$F_{2:0}$	Function
000	$A \& B$ ✓
001	$A B$ ✓
010	$A + B$
011	not used
100	$A \& \sim B$
101	$A \sim B$
110	$A - B$
111	SLT

CONTROL UNIT: MAIN DECODER



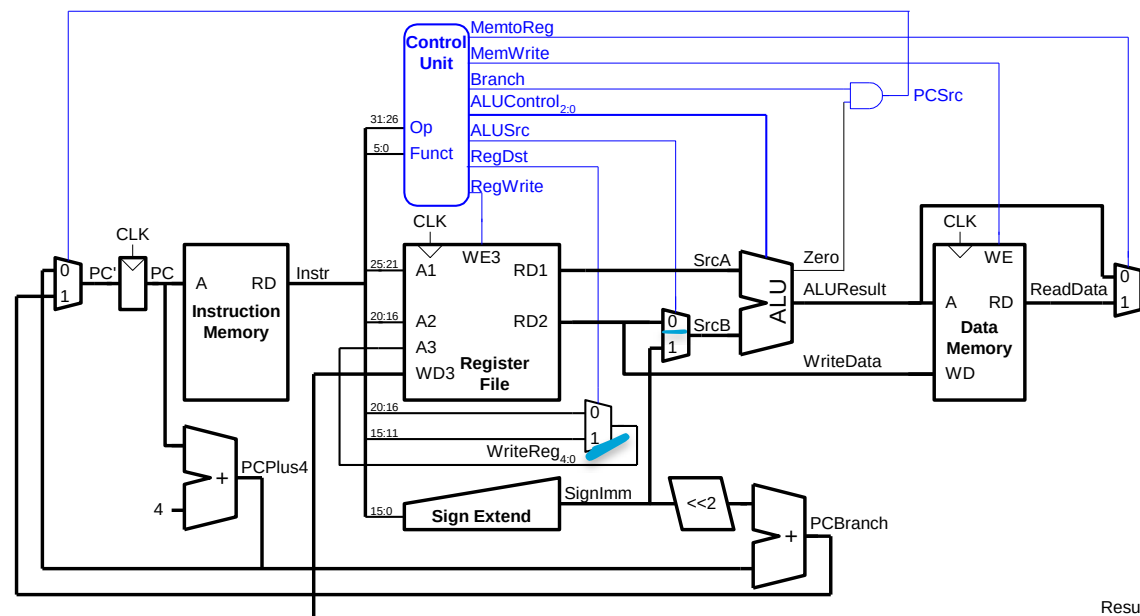
Instruction	Op _{5:0}	RegWrite	RegDst	AluSrc	Branch	MemWrite	MemtoReg	ALUOp _{1:0}
R-type	000000							
lw	100011							
sw	101011							
beq	000100							



CONTROL UNIT: MAIN DECODER



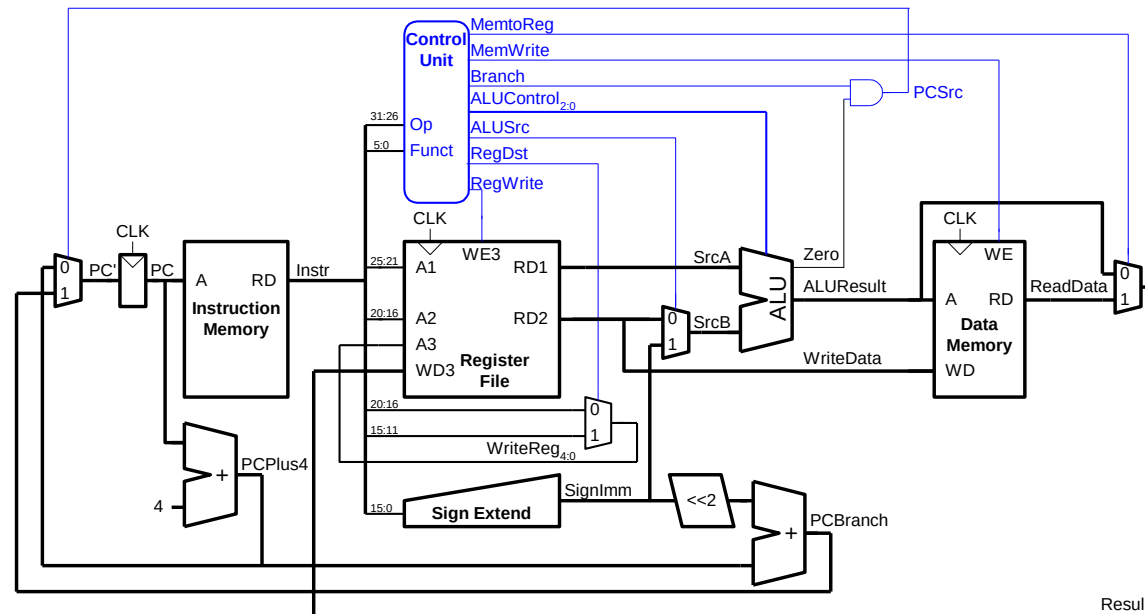
Instruction	Op _{5:0}	RegWrite	RegDst	AluSrc	Branch	MemWrite	MemtoReg	ALUOp _{1:0}
R-type	000000	1✓	1	0✓	<u>0</u>	<u>0</u>	<u>0</u>	<u>10</u>
lw	100011							
sw	101011							
beq	000100							



CONTROL UNIT: MAIN DECODER



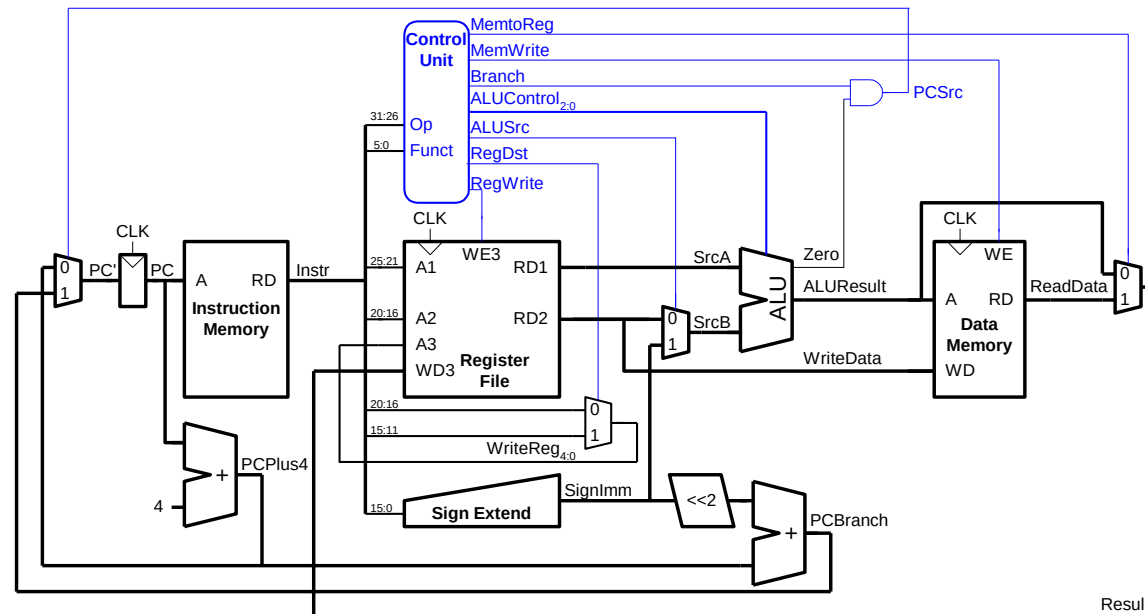
Instruction	Op _{5:0}	RegWrite	RegDst	AluSrc	Branch	MemWrite	MemtoReg	ALUOp _{1:0}
R-type	000000	1	1	0	0	0	0	10
lw	100011	1	0	1	0	0	1	00
sw	101011							
beq	000100							



CONTROL UNIT: MAIN DECODER



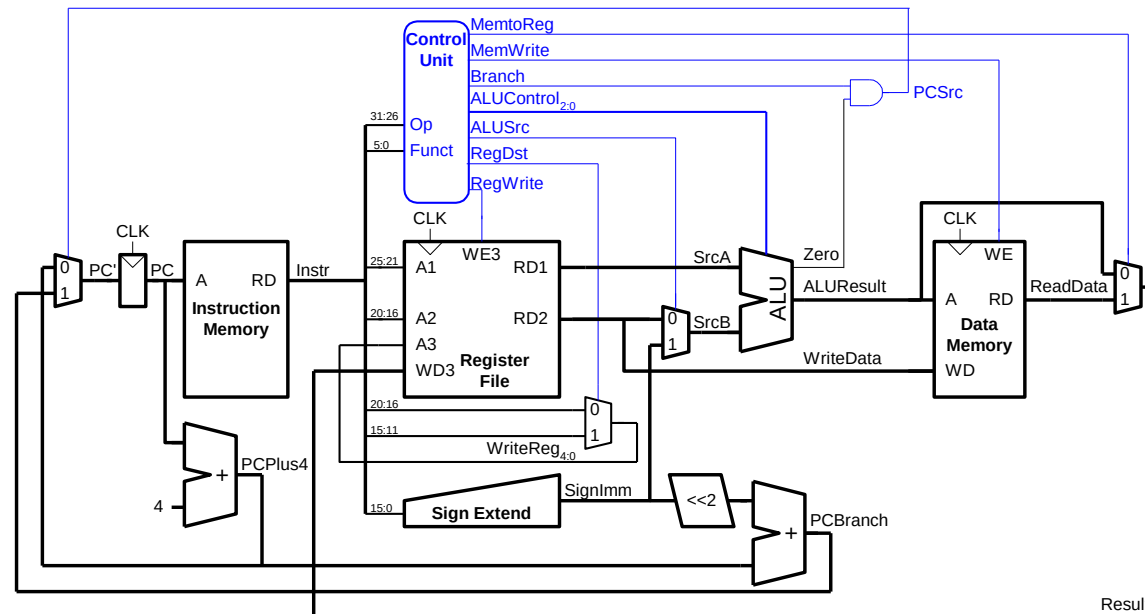
Instruction	Op _{5:0}	RegWrite	RegDst	AluSrc	Branch	MemWrite	MemtoReg	ALUOp _{1:0}
R-type	000000	1	1	0	0	0	0	10
lw	100011	1	0	1	0	0	1	00
sw	101011	0	X	1	0	1	X	00
beq	000100							



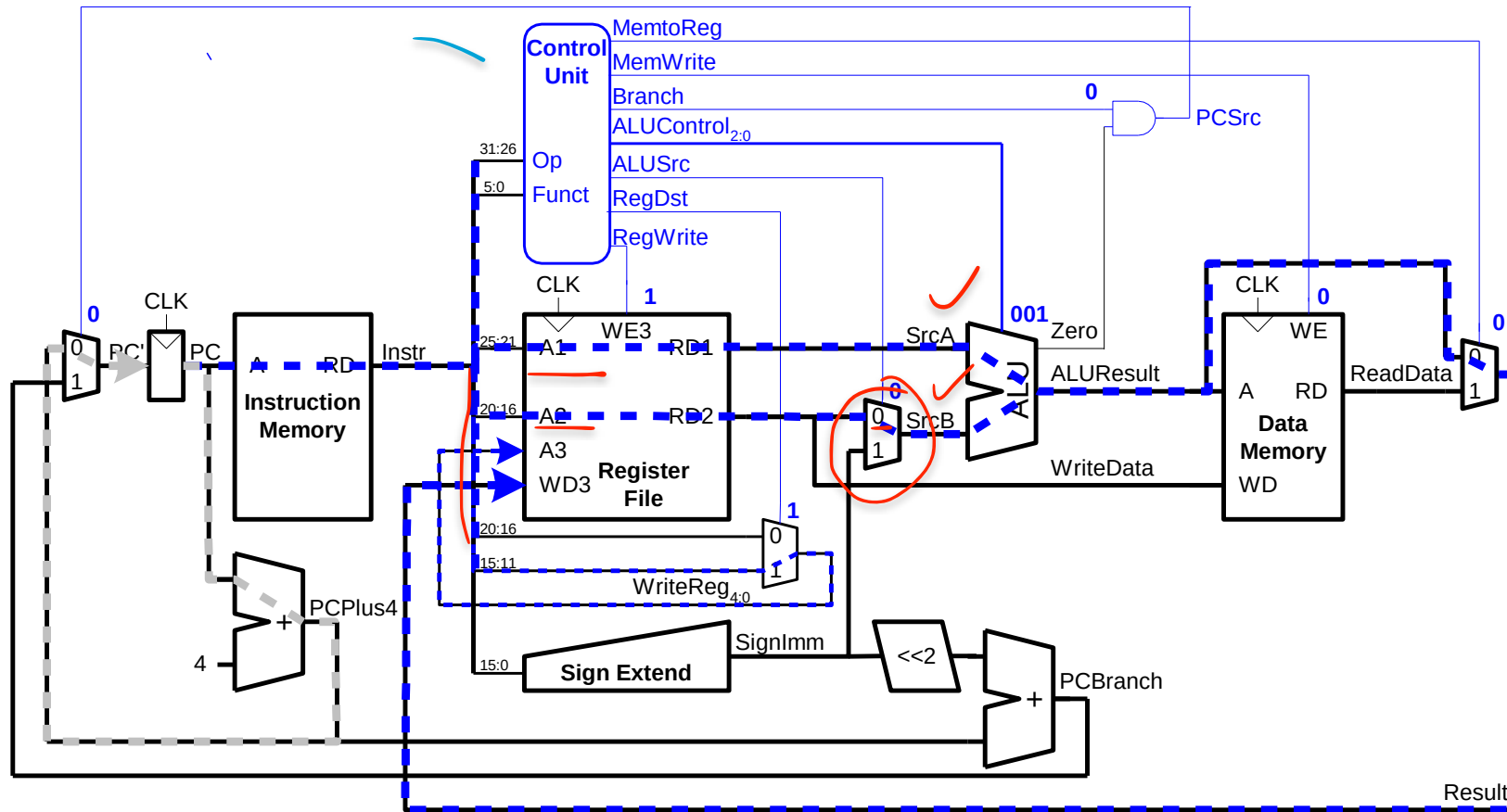
CONTROL UNIT: MAIN DECODER



Instruction	Op _{5:0}	RegWrite	RegDst	AluSrc	Branch	MemWrite	MemtoReg	ALUOp _{1:0}
R-type	000000	1	1	0	0	0	0	10
lw	100011	1	0	1	0	0	1	00
sw	101011	0	X	1	0	1	X	00
beq	000100	0	X	0	1	0	X	01



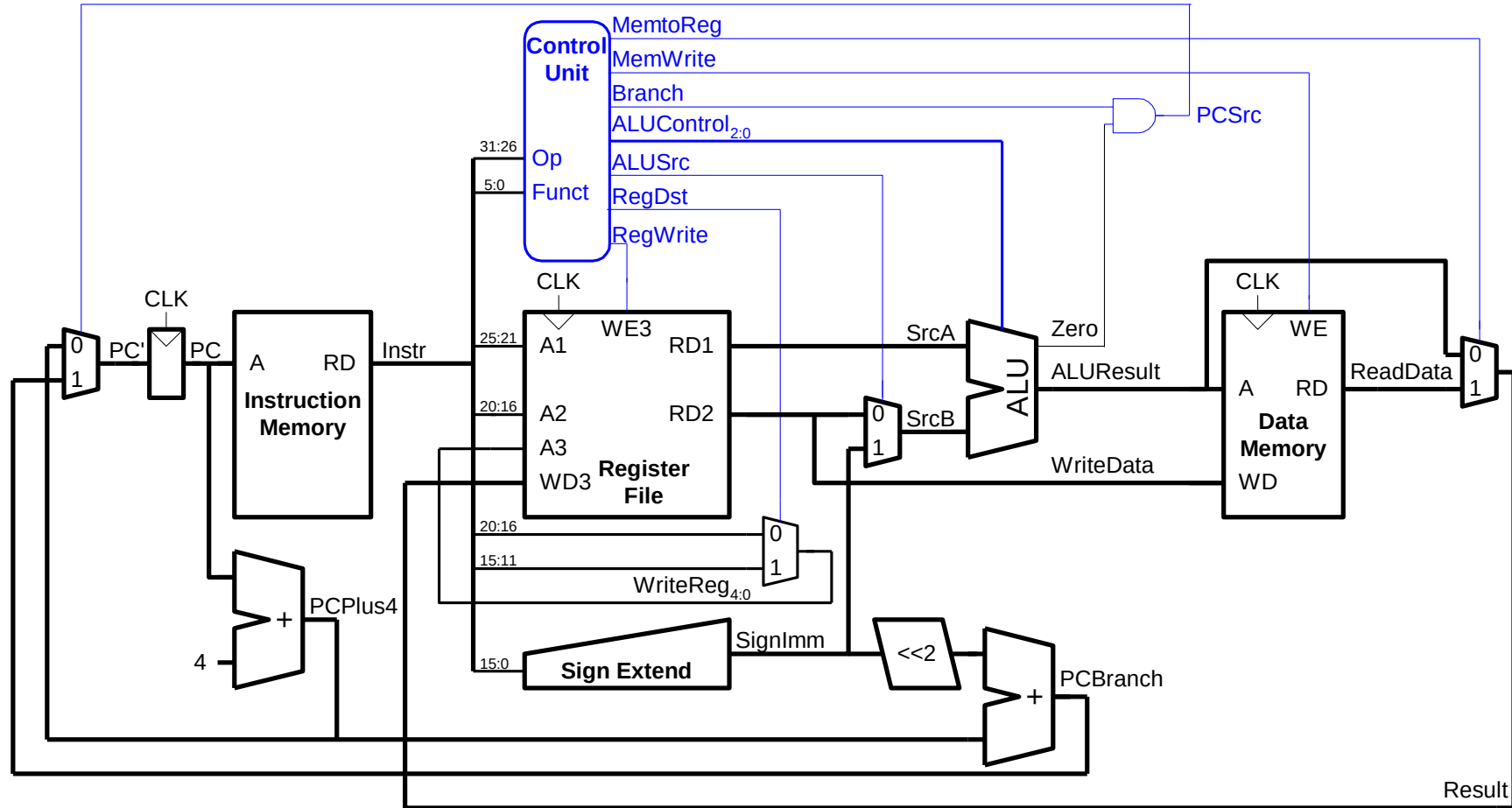
SINGLE-CYCLE DATAPATH EXAMPLE: OR



EXTENDED FUNCTIONALITY: ADDI



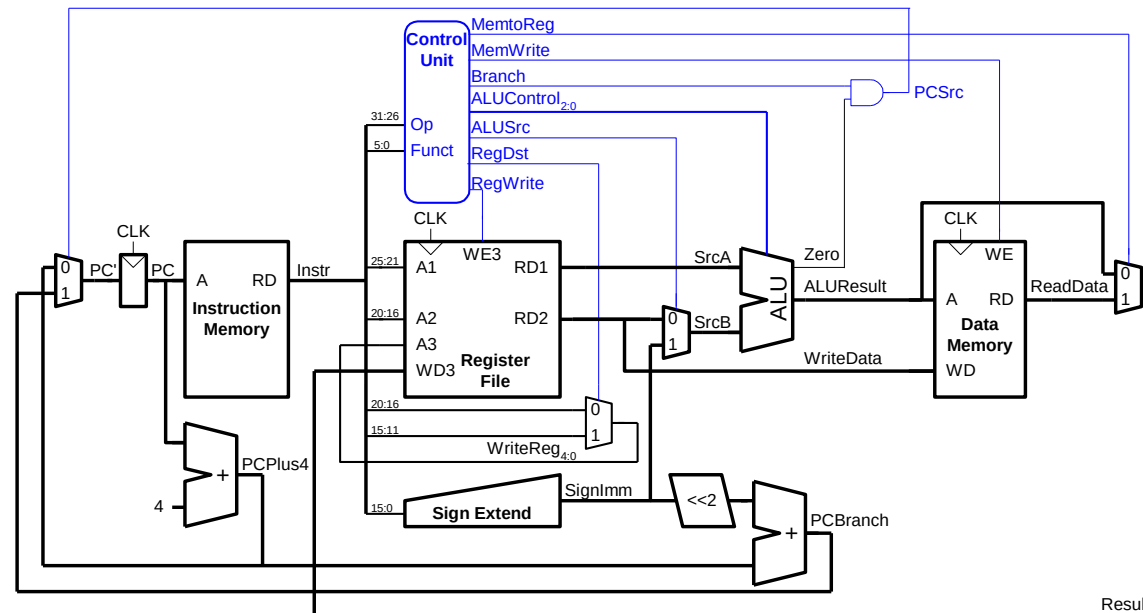
- No change to datapath



CONTROL UNIT: ADDI



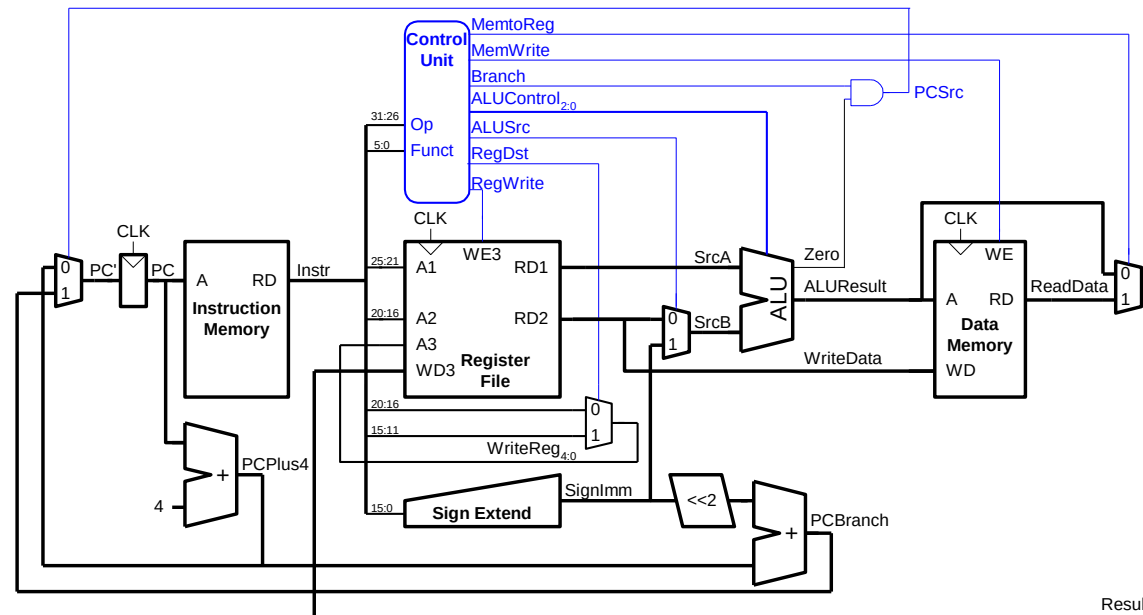
Instruction	Op _{5:0}	RegWrite	RegDst	AluSrc	Branch	MemWrite	MemtoReg	ALUOp _{1:0}
R-type	000000	1	1	0	0	0	0	10
lw	100011	1	0	1	0	0	1	00
sw	101011	0	X	1	0	1	X	00
beq	000100	0	X	0	1	0	X	01
addi	001000							

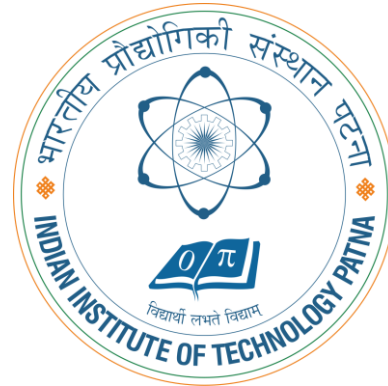


CONTROL UNIT: ADDI



Instruction	Op _{5:0}	RegWrite	RegDst	AluSrc	Branch	MemWrite	MemtoReg	ALUOp _{1:0}
R-type	000000	1	1	0	0	0	0	10
lw	100011	1	0	1	0	0	1	00
sw	101011	0	X	1	0	1	X	00
beq	000100	0	X	0	1	0	X	01
addi	001000	1	0	1	0	0	0	00





THANK YOU!